Introduction to the Course

Autonomous Software Agents

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Outline

- Autonomous Software
  - Agents as new paradigm for software systems development
  - Multi-agent systems
- Objectives of the course
- Structure of the course
- Exam / Project

Turing taught us not to ask whether machines can think, but whether they behave intelligently.”



VOL. LIX. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

BY A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, 'Can machines think?' is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the 'imitation game'. It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart from the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either 'X is A and Y is B' or 'X is B and Y is A'. The interrogator is allowed to put questions to A and B thus:

C: Will X please tell me the length of his or her hair?

Now suppose X is actually A, then A must answer. It is A's

2024 – Billions of AI agents ?

- Billions of AI agents interacting with people: possibly **more AI agents than humans**
- In the future, **every business will have AI agents** will handle:
  - Customer support
  - Sales
  - Communication
  - Personalized interactions



Autonomous Software Agent

A software that can operate autonomously (without any human or other software control):

- Wait a minute ! … any software is like that, right?

  Yes, knowing in advance that cond_1 and battery_NotOK are the only variables that can change the execution

  But, what about obstacle_in_front, should we change the software?

  Also, when cond_1, is it always true that move_right and pick_up are the best actions to do? Are there no alternatives?

```
while(run){
    if(cond_1) {
        move_right()
        pick_up()
    } else {
        move_left()
        mov_on()
    }
    if(battery_NotOK) {
        run = False
    }
}
```

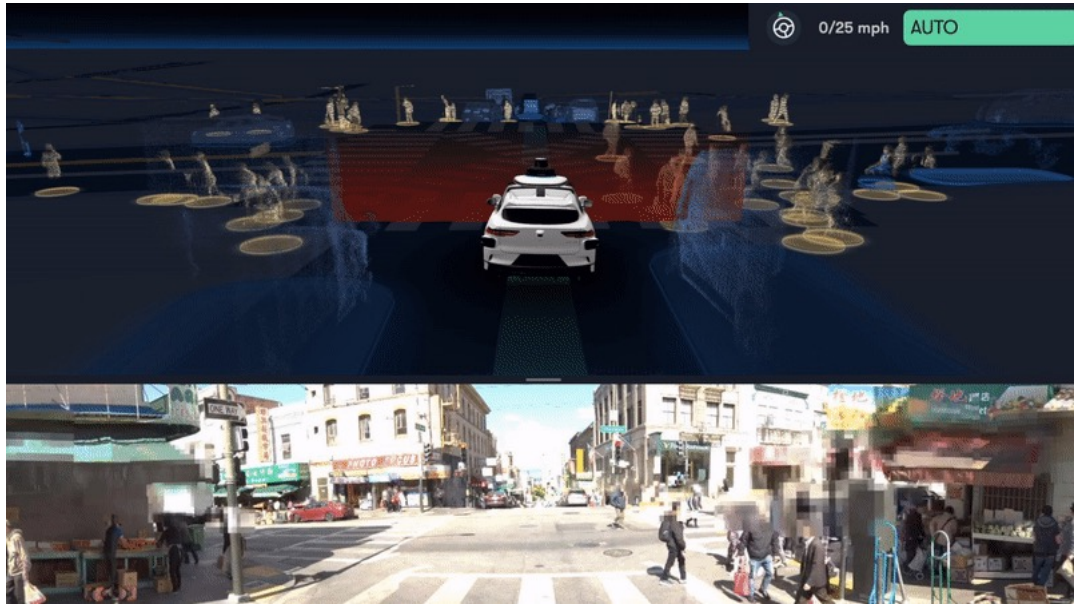
Intelligence and delegation

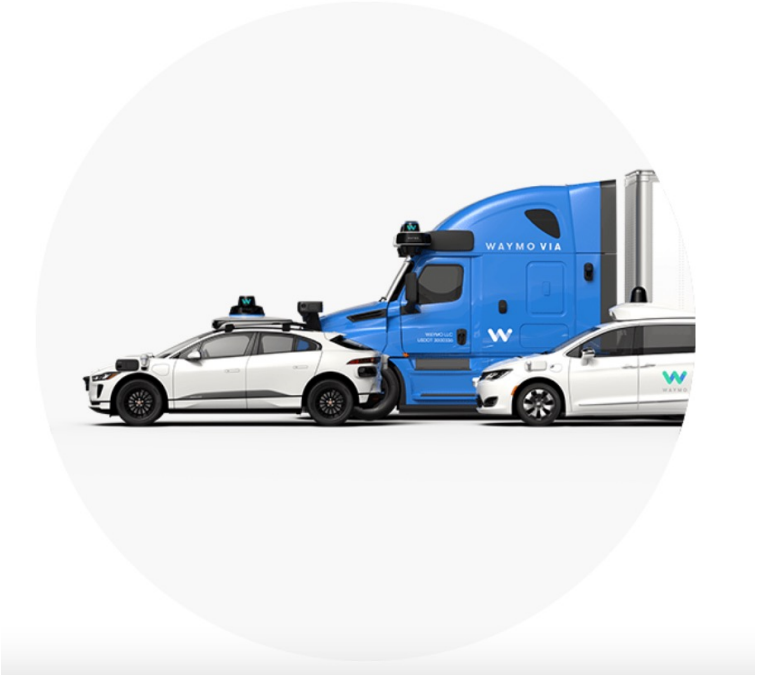
- The complexity of tasks that we are capable of automating and delegating to computers has grown steadily
- Computers are doing more for us – without our intervention
- We are giving control to computers, even in safety critical tasks





Google’s Driverless Car – Waymo Driver





How safe driver is your average robot? Safer than your average American, at least by one measure

- Since 2009, billions of miles driven (millions in public roads) over +13 US states

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The 6 Levels of Vehicle Autonomy

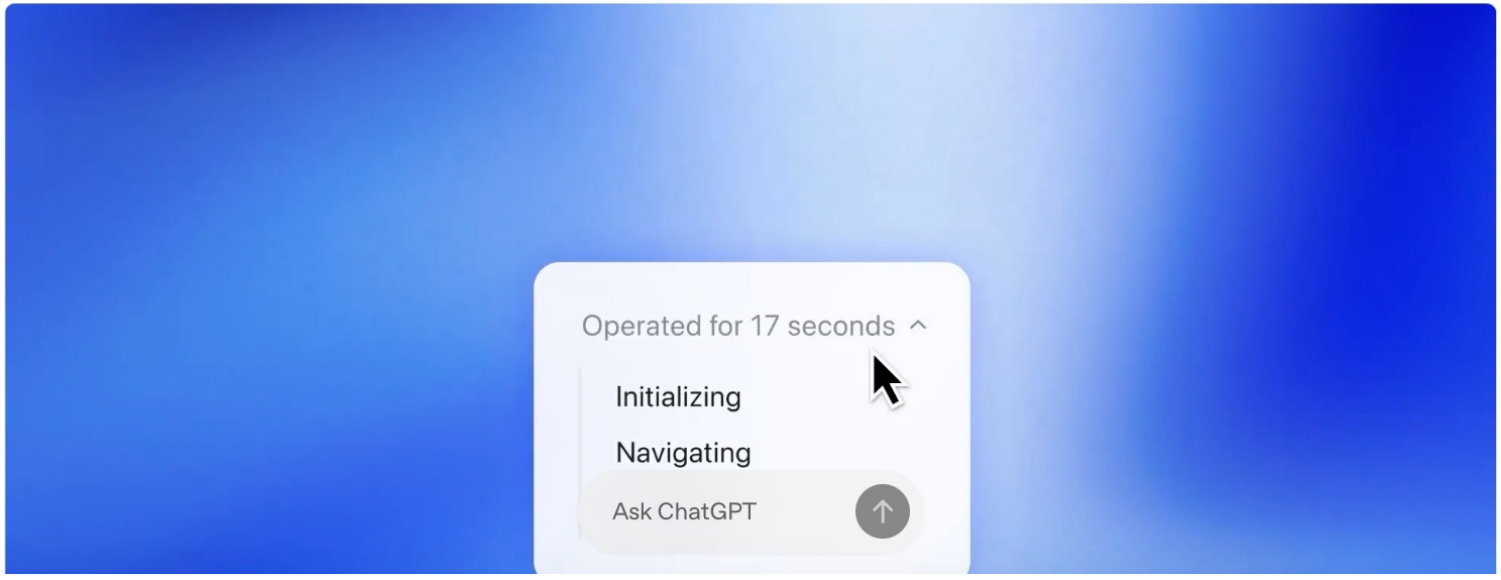
- Level 0 (No Driving Automation)
  - The automated system issues warnings and may momentarily intervene but has no sustained vehicle control
- Level 1 (Driver Assistance)
  - The driver and the automated system share control of the vehicle
- Level 2 (Partial Driving Automation)
  - The automated system takes full control of the vehicle: accelerating, braking, and steering.
- Level 3 (Conditional Driving Automation)
  - The driver can safely turn their attention away from the driving tasks
- Level 4 (High Driving Automation)
  - As level 3, but no driver attention is ever required for safety
- Level 5 (Full Driving Automation)
  - No human intervention is required at all

January 23, 2025 Release

Computer-Using Agent

Powering Operator with Computer-Using Agent, a universal interface for AI to interact with the digital world.

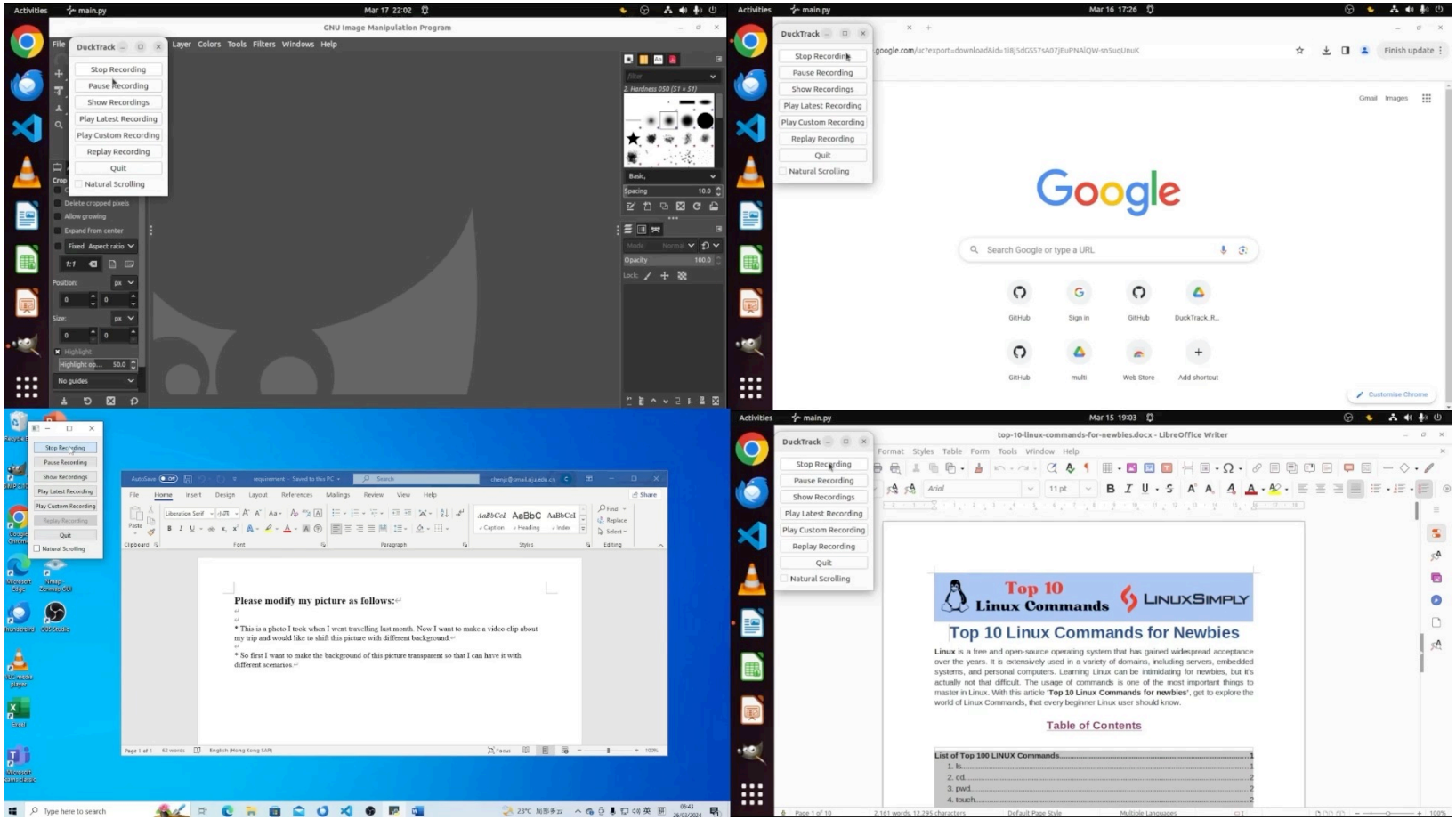
Go to Operator ↗



Computer-Using Agents

- An AI model designed to perform computer tasks using screenshots, cursor, and keyboard
- Combines GPT-*'s vision with advanced reasoning via reinforcement learning
  - Automates tasks like form-filling, travel booking, and content creation
  - Interacts with web browsers similar to human users
  - Pauses for user input on complex tasks (e.g., CAPTCHAs, passwords)
- Collaborative approach ensuring users can guide and intervene
  - Proactive refusals for high-risk actions
  - Confirmation prompts before critical tasks
  - Active monitoring systems to detect threats

OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments



Where does it bring us?



- Delegation and Intelligence imply the need to build computer systems that can act effectively on our behalf
- This implies:
  - The ability of computer systems to act independently
  - The ability of computer systems to act in a way that represents our best interests while interacting with other humans or systems



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Programming progression…



Programming has progressed through:

- machine code;
- assembly language;
- machine-independent programming languages;
- sub-routines;
- procedures & functions;
- abstract data types;
- objects;
- services;

to agents and AI-based systems

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Human Orientation

- Movement away from machine-oriented views of programming toward concepts and metaphors that more closely reflect the way we ourselves understand the world
- Programmers (and users!) relate to the machine differently
- Programmers conceptualize and implement software in terms of higher-level – more human-oriented – abstractions
  - Abstractions to cope with complexity



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Agent, a first definition

An agent is a computer system that is capable of independent actions on behalf of its user or owner (**figuring out what needs** to be done to satisfy design objectives, rather than constantly being told) [Wooldridge&Jennigs - 1995]





Interconnection

- Computer systems today no longer stand alone, but are networked into large distributed systems
  - The internet is an obvious example, but networking is spreading its ever-growing tentacles…
- Since distributed and concurrent systems have become the norm, some researchers are putting forward theoretical models that portray computing as primarily a process of interaction



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Where we are …

- Computing is now about interaction and coordination between distinct piece of software
- More and more independent software able to coordinate with others (including humans)
  - The final objective is obtained as an emergent behaviour of multi-software behaviours
  - Open environment (new piece of software can be added or removed)
  - Multi-objective problem



Amazon warehouse (Video)



Ubiquity

- The continual reduction in cost of computing capability has made it possible to introduce processing power into places and devices that would have once been uneconomic
- As processing capability spreads, sophistication (and intelligence of a sort) becomes ubiquitous











Internet of Things

- The Internet of Things refers to uniquely identifiable objects (things) and their virtual representations in an Internet-like structure
  - If all objects of daily life were equipped with radio tags, they could be identified and inventoried by computers
- Ambient Intelligence (AI) and Autonomous Control (AC) are not part of the original concept of the Internet of Things
  - How to use the Internet of Things for AI and AC?



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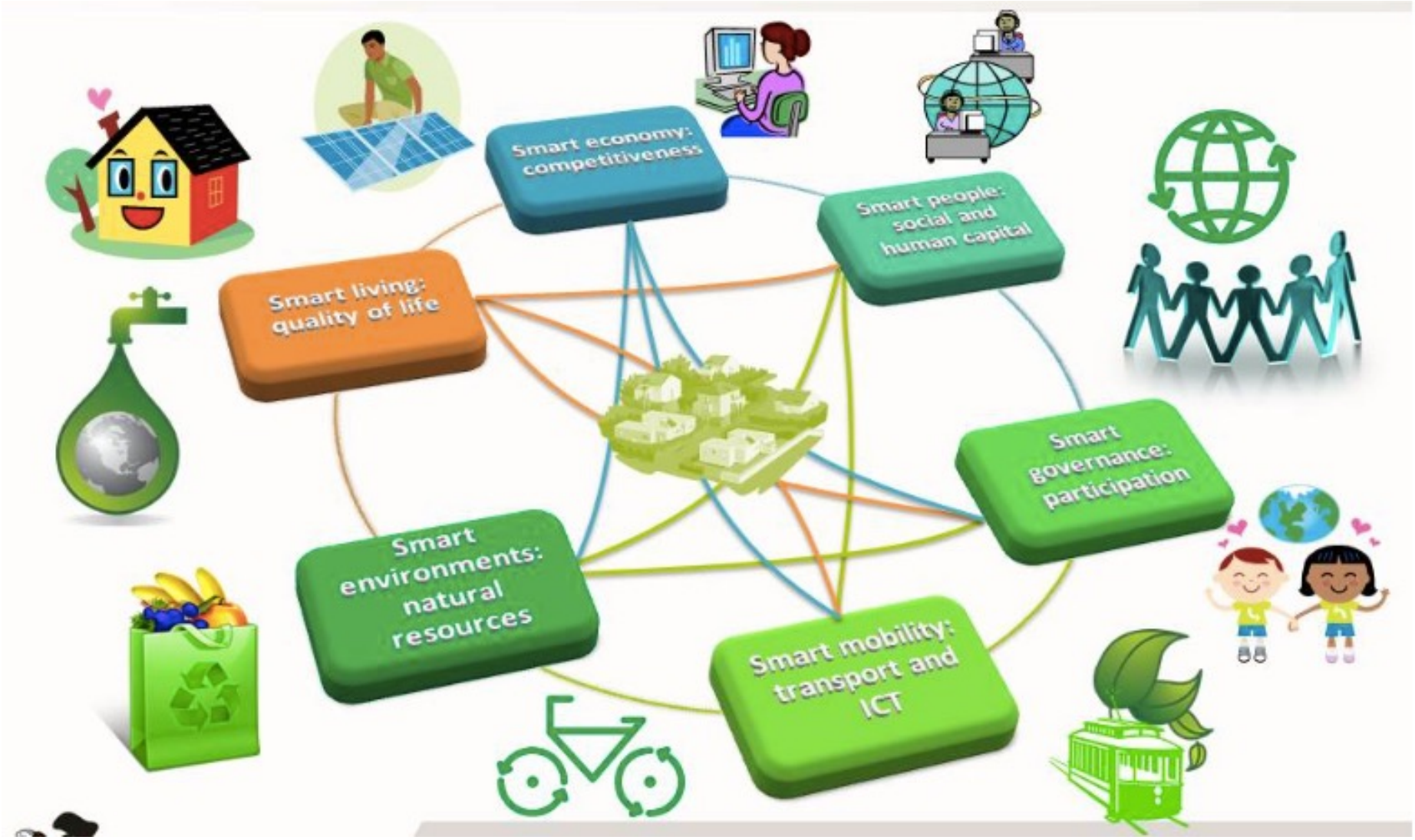
Services, Cloud and Edge computing



- Services are made available to be used and combined together
  - Data storage, software as service, platform as service, infrastructure as service
  - Interaction between different systems is the key issue
  - Quality and security control problems
  - Services availability
  - Run-time composition
  - Flexibility and scalability issues
- Edge computing brings computation and data storage closer to the sources of data

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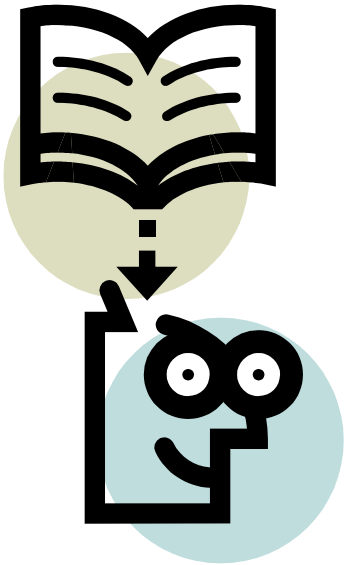
Interconnection: smart city



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Interconnection and distribution

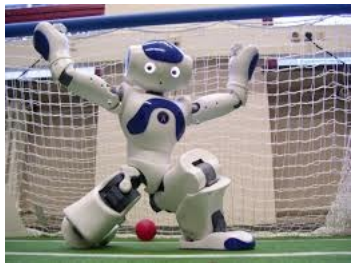
- Interconnection and distribution have become core motifs in Computer Science
- But Interconnection and distribution, coupled with the need for systems to represent our best interests, implies systems that can cooperate and reach agreements (or even compete) with other systems that have different interests (much as we do with other people)



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Examples





Robocup





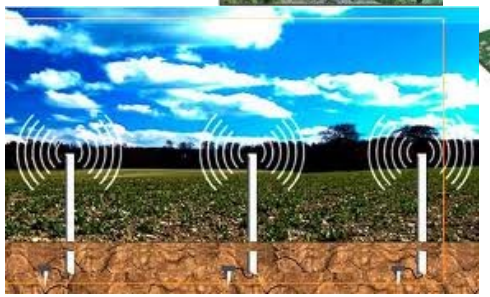






Swarm intelligence







Distributed Sensing

Computer Science expands…

All of these trends have led to the emergence of a new field in Computer Science:

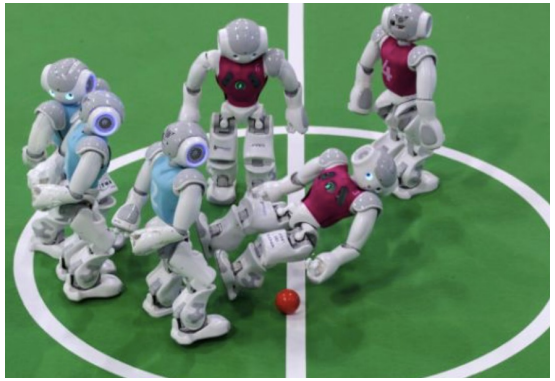
Agent Systems provide us with the means to design and implement interactive computing, whether between machines, or people, or both

[Michael Luck]



Multi-Agent System, a definition

- A multi-agent system is one that consists of a number of agents, which interact with one-another
- In the most general case, agents will be acting on behalf of users with different goals and motivations
- To successfully interact, they will require the ability to cooperate, coordinate, and negotiate with each other, much as people do

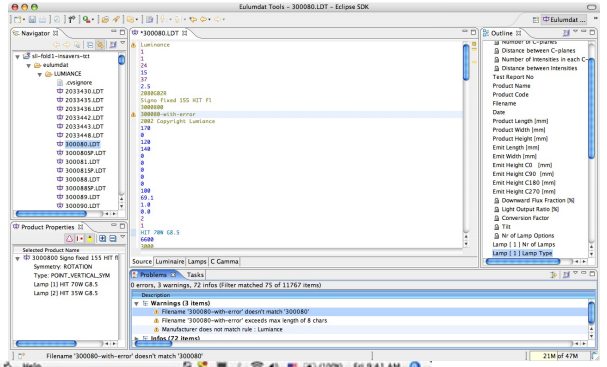


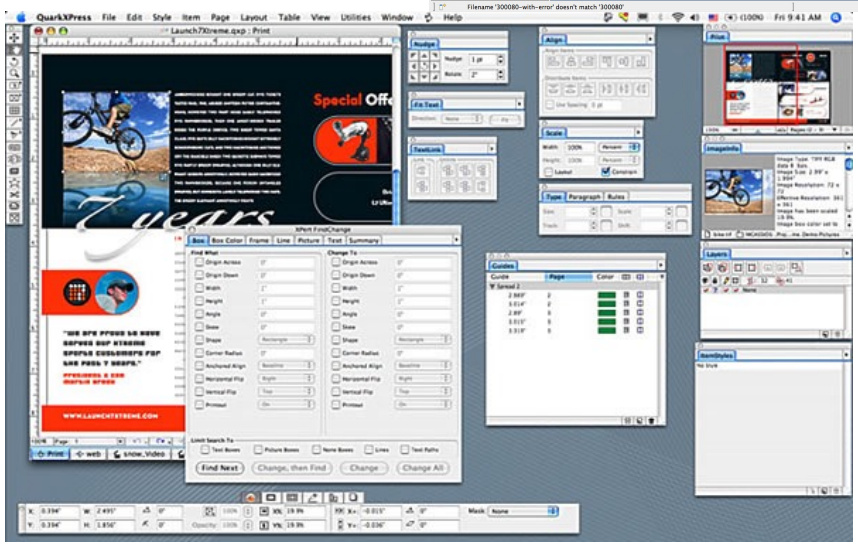


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How can we develop such complex software?

- A good (Agent-Oriented) Software Engineering Methodology
  - Modeling languages
  - Analysis techniques
  - Design Techniques
  - Supporting Tools





Objective of the Course

- The objective of the course is to examine and explore the credentials of agent-based approaches to develop autonomous software and to gain an insight into what agent-oriented software engineering will look like
  - Agents as building blocks of a software system
  - Agent as basic concept to develop a software system
  - Knowledge level Software Engineering
    - Agent concept and its mentalist notions (e.g., goal, plan, and belief) are used along all the software engineering process
  - Multi-Agent Systems
    - Decomposition into sub-agents to cope with complexity
    - Scalability through distributed and coordinated agents
    - Clear separation of roles, goals, and responsibilities
    - Modularization improves: Maintainability, Reusability and Robustness

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Structure of the course

- Very practical course
  - You will learn how to develop both a single and a team of autonomous software systems
- Theory
  - We will explore concepts of autonomous software agents and multi-agent systems, software architectures (including LLM agents), interaction and communication techniques, planning techniques
- Laboratory
  - Tutorial + project development

Tentative Theory content



Introduction

- Introduction to the agent paradigm and multi-agent systems
- Applications and problems

Interaction and cooperation

- Types of interaction
- Forms of Cooperation
- Methods of cooperation
- Organization and cooperation

Communication

- Aspects of communication
- Speech acts
- Conversation
- KQML/ACL

Collaboration and distributed tasks

- Models of tasks allocation
- Centralized allocation
- Decentralized allocation
- Emergent allocation

The BDI Architecture

- The agent control loop
- The deliberation process

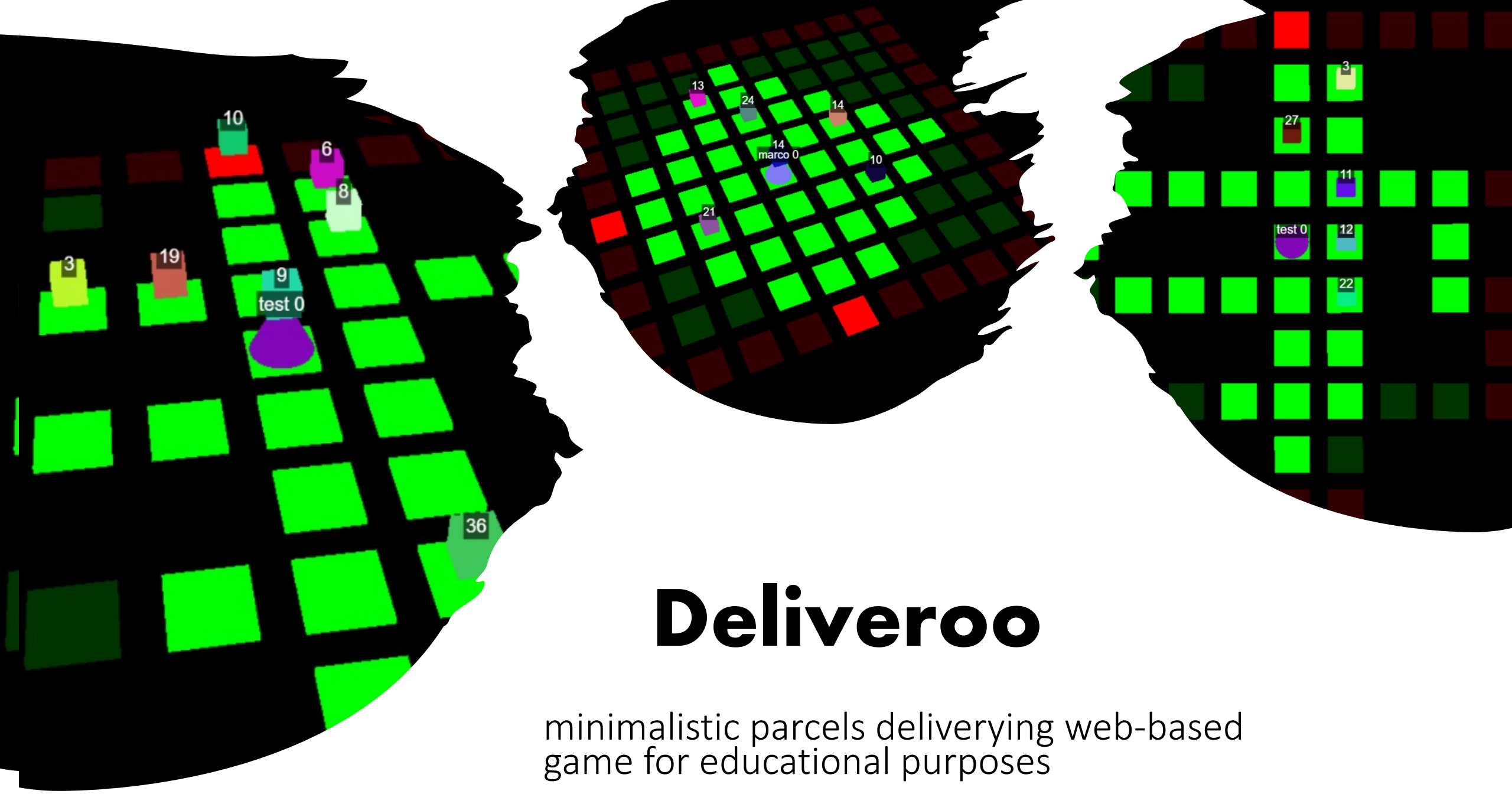
LLM agents

- Principles and techniques
- LLM reasoning and planning

Planning

- Exploring the space of alternatives
- Plannnig in the agent control loop

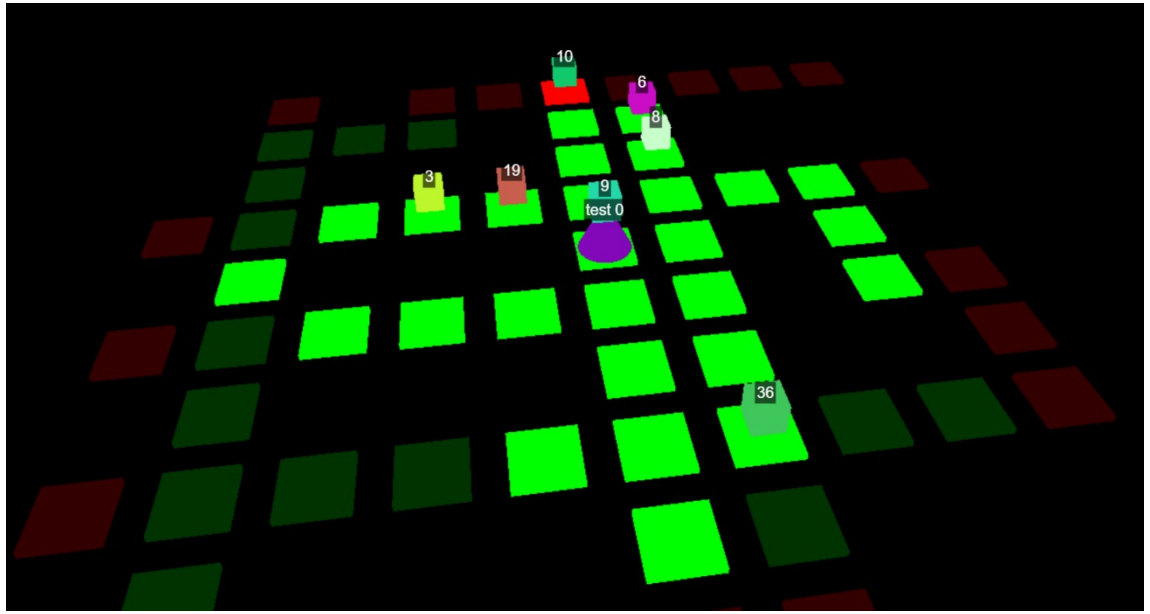
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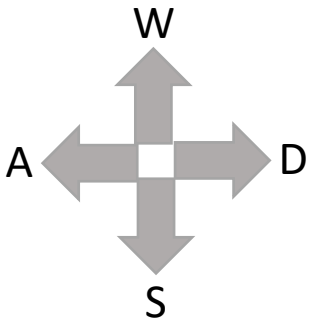
Deliveroo

minimalistic parcels deliverying web-based game for educational purposes

Deliveroo – try it!



https://deliveroojs.azurewebsites.net/

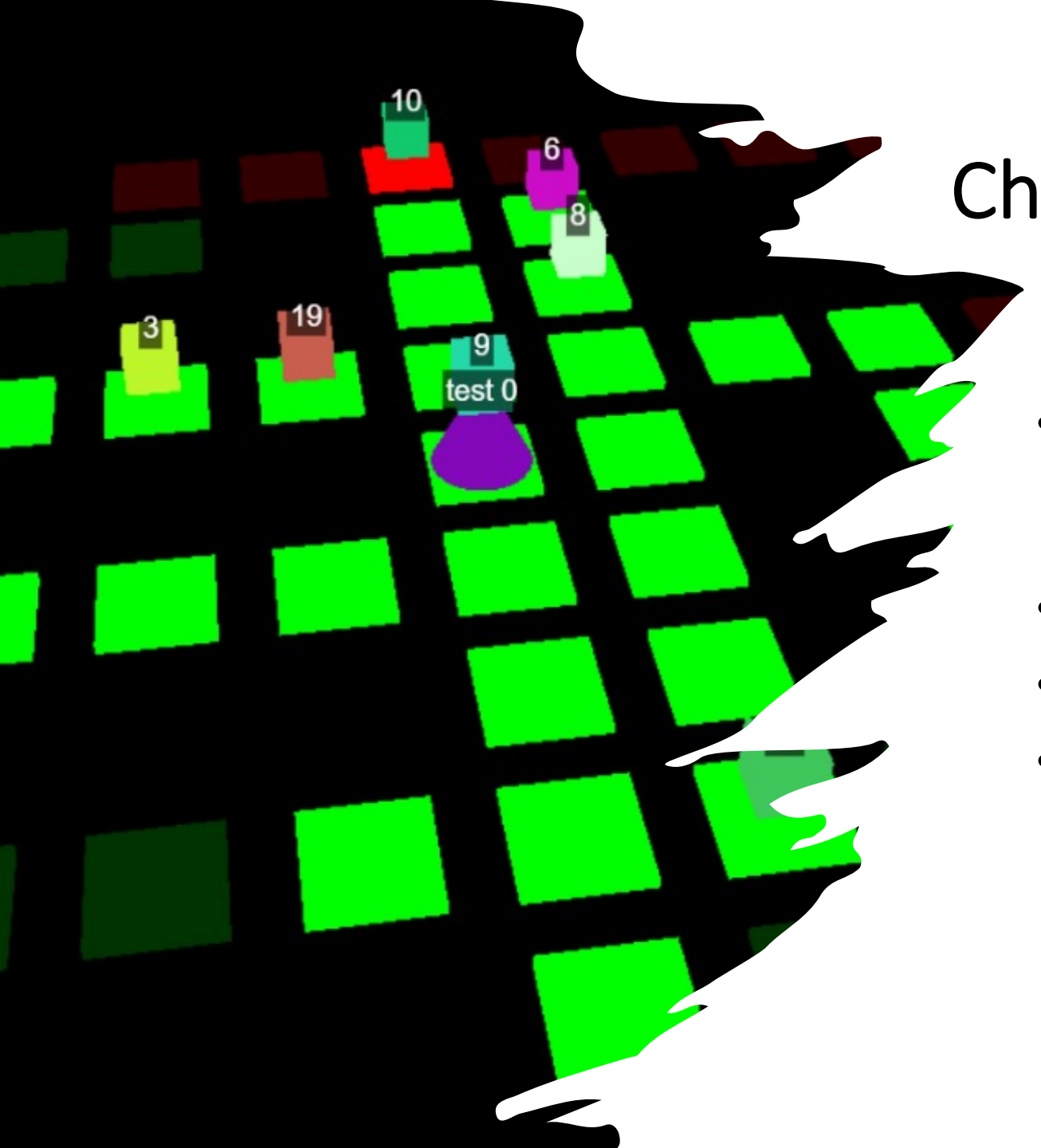


Q: pick_up
E: put_down

The objective of the project

- Develop an autonomous software that will play on your behalf
  - earning points by collecting as many parcels as possible and delivering them in the delivery zone
  - **First part**: using a BDI architecture
    - Sensing the environment and managing Beliefs
    - Deliberating Intentions
    - Select plans from a library
    - Using an external planner component
    - Execute a plan (actions)
    - Defining strategies and behaviours
    - Replanning and redeliberating
  - **Second part**: the user (you) can give instructions to the agent in textual form
    - Use a LLM-agent to interact with user and control the BDI-agent

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Challenges and turnments

- In order to validate your ASA we will organize tournments where each group will challenge the other goups.
- Single agent tournement
- Human ion the loop tournment
- Different level of complexity



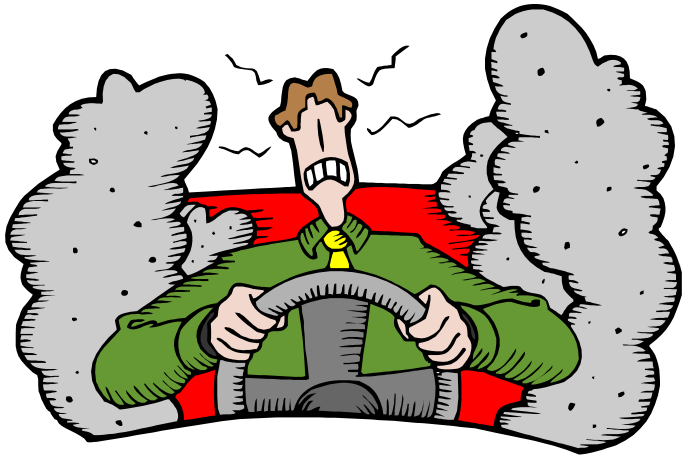






Laboratory

- Hands-on with exercises
- Node.js – javascript
- Planning
  - PDDL
- LLM agents
- Project development



→ (Take your laptop with you)

Exam and prerequisites



Project: a scenario will be provided in a couple of weeks

- Groups of 2 students
- First part: single BDI-agent
- Second part: team of two collaborative agents (LLM + BDI agent)

- Exam
  - Deliverable
    - Code + Report (max 10 pages)
  - Project presentation
- Prerequisite for the course: experience in programming
    - Better if you know Javascript (we will provide a Tutorial)

Books and Material



- Given the fact we will try and provide a broad coverage of various topics seems unrealistic an approach in which all the textbooks are adopted.

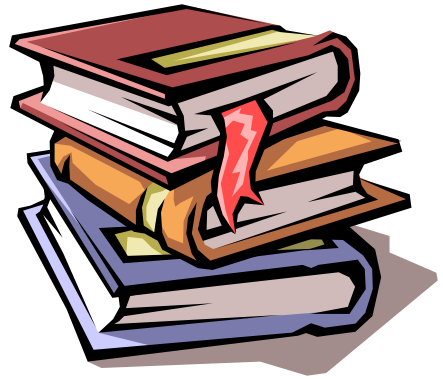
Therefore:

- We encourage the use of slides, papers, (on-line) material, etc.
- We provide a list of textbooks of which only parts are needed (no need to get all of them, though!)

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Books…

- *Jacques Ferber. "Multi-Agent System: An Introduction to Distributed Artificial Intelligence," Addison Wesley Longman, 1999.*
- *Michael Wooldridge. "An Introduction to Multiagent Systems," John Wiley & Sons, 2002 - see subsequent editions.*
- *Brian Henderson-Sellers and Paolo Giorgini. "Agent-oriented Methodologies," Idea Group Inc, 2004.*
- *Chip Huyen. AI Engineering: Building Applications with Foundation Models. O'Reilly, 2024.*
- *Edward Chang. Multi-LLM Agent Collaborative Intelligence: The Path to ArtificialvGeneral Intelligence, ACM Books, 2026.*



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Next lectures

- Tuesday 11.30 -13.30 (A110)
- Wednesday 9.30-11.30 (A110)

Lectures will be in presence, but we will stream (zoom) and record them

All info and material is Moodle

- Zoom link for the lectures
- Slides + Video
- GitHub
- … and much more

Contacts



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